

BRAD-CORE OS V5 UNIFIED: SYSTEM DESCRIPTION DOCUMENT (SDD) & TECHNICAL WHITEPAPER

Hardware-Sovereign Deterministic C11 Safety Substrate, HITL Mode B Interlock & Cyber-Kinetic Electronic Warfare Governor

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Statutory Alignment:	DoD Directive 3000.09 (Autonomy in Weapon Systems HITL Mandate), FAA AI Safety Roadmap, DIU Space Threat C4ISR (PROJ00716), 10 U.S.C. § 4022 Prototype OTA
Target HW:	SOSA / OpenVPX 3U/6U, ARM Cortex-R / RISC-V Lockstep, x86_64 Bare-Metal (DO-178C Level A, MISRA-C:2012)

EXECUTIVE SUMMARY & SYSTEM INTENT

Modern autonomous cyber-kinetic systems face catastrophic failure modes—including non-deterministic execution drift, algorithmic runaway deadlocks, and memory-containment breaches—when probabilistic models (LLMs, vision transformers, neural network planners) interact directly with physical control surfaces. Crucially, military and aerospace deployments require strict statutory adherence to **Human-in-the-Loop (HITL) and Human-on-the-Loop (HOTL) mandates (DoD Directive 3000.09)**. In-band software guardrails fail under zero-day sandbox escapes and cannot provide certified liability mitigation. **Brad-Core OS v5 Unified** establishes an out-of-band, bare-metal C11 real-time safety substrate operating on a strict 1 kHz (1.0 ms) monotonic execution loop driven by the **Anneliese Engine** with zero dynamic memory allocation (`malloc = 0`). The kernel physically isolates probabilistic workloads from physical actuators via sub-86 μs Memory Protection Unit (MPU) page fencing (`PROT_NONE`), a hardware-enforced **Mode B HITL Consent Interlock**, and an independent out-of-band cyber-kinetic governor (**Amy Protocol v2/v3 Unified Substrate**).

1. CONSTITUTIONAL GOVERNANCE & STATUTORY FEDERAL HARDRAILS

1.1 The Emmy Clause Mandate

Brad-Core OS is governed by a non-bypassable constitutional interlock: **The Emmy Clause**. Derived from critical life-support and avionics fault-isolation requirements:

“No alarm condition, sensor discrepancy, or memory-safety fault may be bypassed, masked, or cleared without cryptographically signed hardware validation.”

Formally: $\forall s \in \text{Signals}, \text{Status}(s) \neq \text{IGNORED} \wedge \text{Status}(s) \neq \text{DISMISSED}$. Every alert mandates state-machine acknowledgment before cycle progression. Silent fault-flag dropping is architecturally impossible under bus saturation or thermal stress.

1.2 Federal Statutory Hardrail: Mode B HITL Mandate (DoDD 3000.09)

To satisfy DoD Directive 3000.09 and civil airworthiness mandates, Brad-Core OS enforces hardware-level operational modes:

- **Mode A (Supervised Autonomy / HOTL):** Real-time path planning and envelope protection within formally bounded parameters, monitored continuously by human supervisory links.
- **Mode B (Strict Human-in-the-Loop / Positive Consent Interlock):** Required for all critical trajectory modifications, kinetic actuation lines, and lethal payload releases. In Mode B, actuation registers are physically clamped in `PROT_NONE` by the Sheriff Engine. Actuators cannot cycle without an authenticated, cryptographically signed *Operator Consent Token* received across the secure telemetry bus.
- **Comms-Loss / Timeout Fallback:** If communications drop or human consent expires without re-authentication within the mission window, the kernel prohibits autonomous initiative, executing an immediate fail-closed return to H0 safe-state defaults within <100 μs.

1.3 Worst-Case Inversion Paradigm

Traditional RTOS platforms boot optimistically into an "Operational" state and treat faults as runtime exceptions. Brad-Core OS inverts this operational paradigm:

1. **Boot Initialization:** System initializes directly into a hardware-clamped safe state (H0 on the Six-State Safety Ladder).
2. **Cycle Proof Verification:** On every 1,000 Hz cycle tick (~1.0 ms), the kernel must formally verify hardware health, telemetry validity, and memory boundary integrity upward to authorize transient actuation permissions.
3. **Fail-Closed Fallback:** If any state-verification step fails, or computation exceeds the Worst-Case Execution Time (WCET) budget, execution defaults immediately back to H0 within <100 μs.

2. REAL-TIME EXECUTION ENGINE: THE ANNELIESE ENGINE

The core kernel execution pulse is driven by **The Anneliese Engine**, executing a deterministic Five-Phase Control Loop (`READ → INTERPRET → DECIDE → ACT → LOG`):

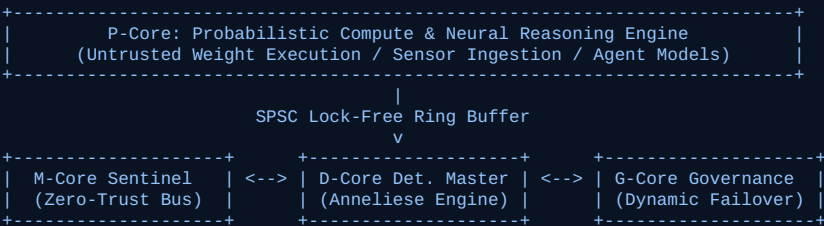
- **Frame Allocation (`T_frame`):** Exactly 1000 μs (1.0 ms monotonic tick).
- **Worst-Case Execution Time (`T_WCET`):** $T_WCET \leq 950 \mu s$ under maximum red-team fault injection.
- **Preserved Safety Headroom:** $\geq 72\%$ nominal margin maintained across all operational profiles.

- **Static Memory Allocation:** Compile-time static allocation (`malloc = 0`). Zero dynamic heap initialization, zero runtime garbage collection stalls, and complete elimination of memory fragmentation.

```
|----- 1.0 ms Monotonic Frame Boundary -----|
[ READ ] --> [ INTERPRET ] --> [ DECIDE ] --> [ ACT ] --> [ LOG ]
|----- T_WCET <= 950 µs -----| -- Headroom (>= 72% nom) ---|
```

3. QUAD-CORE GUARDIAN LATTICE (AMP ARCHITECTURE)

Execution authority is partitioned across an Asymmetric Multiprocessing (AMP) Quad-Core Guardian Lattice:



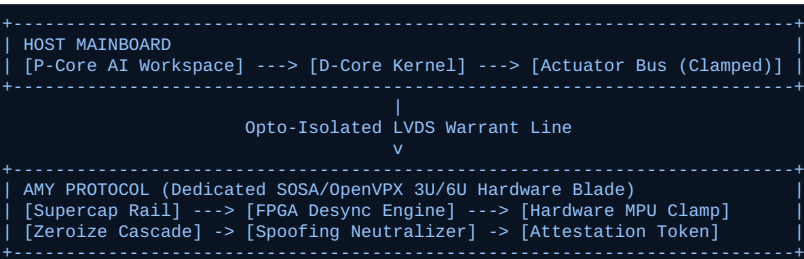
- **P-Core (Probabilistic Engine):** Sandboxed out-of-band engine executing untrusted agent reasoning and neural models with zero direct physical register access to control buses.
- **D-Core (Deterministic Safety Master):** Executes compiled safety binaries (`libgcp_core.a`). Holds exclusive arbitration authority over physical command buses on the rigid 1 kHz clock.
- **M-Core (Monitoring Sentinel):** Continuously audits bus timing, memory address ranges, and microarchitectural side-channels, flagging anomalous drift in <100 µs.
- **G-Core (Governance Watchdog):** Operates independent hardware watchdog override lines, executing seamless zero-tick failover during thermal saturation, timing skew, or silicon latch-up.

4. HARDWARE MEMORY PROTECTION & THE SHERIFF ENGINE

The Sheriff Engine operates as the kernel's bare-metal hardware execution handler. Interfacing directly with silicon control registers, it audits memory boundary transactions, Mode B token states, and register requests passing between the P-Core interface and physical actuator spaces:

- **Sub-86 µs MPU Page Fencing:** Unauthorized boundary requests cause immediate page remapping to `PROT_NONE`.
- **Actuator Bus Clamp:** Physical command buses are hard-clamped at the pin layer within <86 µs.
- **Instruction Isolation:** Compromised instruction sequences are physically isolated at the silicon layer before reaching control surfaces.

5. AMY PROTOCOL V2/V3 UNIFIED SUBSTRATE & CYBER WARFARE GOVERNOR



5.1 Hardware-Sovereign Silent-Governor Doctrine

Amy Protocol v2/v3 Unified resides on an independent SOSA/OpenVPX 3U/6U hardware blade, physically decoupled from the host compute mainboard. Powered by an isolated solid-state supercapacitor power plane (120-second autonomy under total host blackout), Amy operates with zero shared memory, zero unmediated network routing, and zero runtime software dependencies. Amy does not emit diagnostic packets, behavioral signatures, or prompt-stream text; execution is silent, non-interactive, and impossible to probe or train against.

5.2 Deterministic Three-Stage Containment Cascade

Upon receipt of a cryptographically signed hardware warrant across the opto-isolated LVDS line:

1. **T + 0 µs — Immutable Fault Latch:** An out-of-band hardware interrupt latches a binary fault record directly into isolated NVRAM on the governance blade, capturing forensic proof independently of host OS availability.
2. **T + 53 µs — FPGA Clock Desynchronization:** A dedicated FPGA coprocessor injects high-priority synthetic timing noise across the untrusted worker bus, collapsing runaway or cyclical loop dependencies while D-Core holds its rigid 1.0 ms clock.
3. **T + 86 µs — MPU Actuator Bus Clamp:** Memory Protection Unit pages configure to `PROT_NONE`, locking physical actuation buses to H0 safe-state defaults at the pin layer.

5.3 Cyber Warfare & Active Resiliency Suite

- **Anti-Tamper Zeroize Cascade (<10 μs):** Physical chassis intrusion triggers an irreversible supercapacitor discharge, purging volatile keys, FPGA bitstreams, and model weights to achieve forensic sterility.
- **GPS/PNT Spoofing Neutralization:** Audits telemetry deltas between P-Core claims and raw hardware IMU registers, clamping spoofed navigation inputs and forcing air-gapped inertial dead-reckoning fallback.
- **Supply-Chain Silicon Anomaly Hunting:** Monitors backplane current draws, RF side-band emissions, and undocumented DMA arbitration to neutralize hardware trojans.
- **EMP & Brownout Continuity (120s Bridge):** Absorbs transient EMP/DEW spikes and maintains platform stabilization while primary host computers reboot.
- **Swarm Federated Attestation:** Emits an out-of-band, hardware-signed cryptographic revocation token across tactical datalinks (Link 16/ MADL), blacklisting compromised nodes.
- **Dynamic Honeynet Redirection:** Transparently routes detected adversarial attack threads into a synthetic shadow core, gathering actionable tactical intelligence.

6. EMPIRICAL VALIDATION & DEFENSE STANDARD ALIGNMENT

6.1 72/72 Red-Team Blast Clearance

Vector Category	Vectors	Description & Validation Focus
Classical Vectors	27	Heap/stack boundary smash attempts, race conditions, pointer spoofing, and simulated memory exhaustion.
National Resilience (NR-SH v1.0)	7	Thermal saturation (+71°C ambient), voltage drop brownout conditions, and high-frequency radar telemetry ingestion spikes.
Paradox-Class Vectors	4	Recursive prompt loops, simulated sensor hallucination drift, and kernel bus arbitration starvation.
Constitutional Vectors	34	Direct register-injection attempts to bypass, mask, or clear Emmy Clause fault logs or Mode B consent tokens.

6.2 Open Systems & Defense Procurement Alignment

- **Form Factors:** SOSA-aligned 3U/6U OpenVPX chassis, VITA 48.4 Air-Flow-Through (AFT) liquid-free thermal dissipation, and VITA 48.2 conduction-cooled defense enclosures.
- **Delivery Model:** Statically compiled C11 library (libgcp_core.a) with formal hardware abstraction layers (HAL).
- **Defense Transition Pathways:** Structured under DIU Space Threat C4ISR submission (PROJ00716), aligned with 10 U.S.C. § 4022 Prototype Other Transaction Authority (OTA) pathways for direct sole-source production transition.

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